

Value of the block rearrangements of the alternating harmonic series

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1 Facts we are borrowing

I am not proving these 2, just borrowing as known properties. First can be shown by arithmetic manipulation, and second can be shown from the digamma I guess.

Fact 1 (odd and even part of a harmonic sequence). For $m \in \mathbb{N}$ let $H_m = \sum_{k=1}^m \frac{1}{k}$. Then

$$\sum_{k=1}^m \frac{1}{2k} = \frac{1}{2}H_m, \quad \sum_{k=1}^m \frac{1}{2k-1} = H_{2m} - \frac{1}{2}H_m.$$

Fact 2 (Euler–Mascheroni constant). γ is defined by

$$\gamma = \lim_{m \rightarrow \infty} (H_m - \ln m).$$

Set

$$\varepsilon_m := H_m - \ln m - \gamma, \quad m \in \mathbb{N},$$

so that

$$H_m = \ln m + \gamma + \varepsilon_m \quad \forall m \in \mathbb{N}, \quad (1)$$

and so, by the definition of convergence of this,

$$\forall \epsilon > 0 \quad \exists M \in \mathbb{N} \quad \forall m \geq M : |\varepsilon_m| < \epsilon. \quad (2)$$

2 Defining the series

Let $a_j = \frac{(-1)^{j+1}}{j}$ for $j \in \mathbb{N}$. Its positive terms, in their natural order, are

$$p_k = \frac{1}{2k-1}, \quad k \in \mathbb{N},$$

and its negative terms, in their natural order, are

$$q_k = -\frac{1}{2k}, \quad k \in \mathbb{N}.$$

Now for $(P, Q) \in \mathbb{N} \times \mathbb{N}$. let's define $F(P, Q)$ to be the infinite sequence built by taking P positive terms, then Q negative terms, then the next P +ve terms, then the next Q -ve terms, and so on.

Formally: for $i \in \mathbb{N}$, by the division algorithm there are unique $n \in \mathbb{Z}_{\geq 0}$ and $r \in \mathbb{Z}$ with

$$i-1 = n(P+Q) + r, \quad 0 \leq r < P+Q.$$

Call $n =: \beta(i)$ the block of i and $r =: \rho(i)$ the position of i inside its block. Define the i -th term of $F(P, Q)$ by

$$b_i^{(P, Q)} = \begin{cases} p_{Pn+r+1}, & 0 \leq r < P, \\ q_{Qn+r-P+1}, & P \leq r < P+Q, \end{cases} \quad n = \beta(i), r = \rho(i). \quad (3)$$

Lemma 1. *Every p_k ($k \in \mathbb{N}$) and every q_k ($k \in \mathbb{N}$) occurs exactly once in the sequence $(b_i^{(P, Q)})_{i \in \mathbb{N}}$. Hence $F(P, Q)$ is a rearrangement of $(a_j)_{j \in \mathbb{N}}$.*

Proof. Let $\mathcal{I}^+ = \{i \in \mathbb{N} : 0 \leq \rho(i) < P\}$ and $\mathcal{I}^- = \{i \in \mathbb{N} : P \leq \rho(i) < P+Q\}$. These partition $\mathbb{N}$, and by (3) the terms indexed by $\mathcal{I}^+$ are exactly the positive ones and those indexed by $\mathcal{I}^-$ the negative ones.

Consider $\varphi : \mathcal{I}^+ \rightarrow \mathbb{N}$, $\varphi(i) = P\beta(i) + \rho(i) + 1$.

Surjective. Given $k \in \mathbb{N}$, write $k-1 = Pn+r$ with $n = \lfloor (k-1)/P \rfloor \geq 0$ and $0 \leq r < P$ (division algorithm). Put $i = n(P+Q) + r + 1$. Since $0 \leq r < P \leq P+Q-1$, uniqueness in the division algorithm gives $\beta(i) = n$ and $\rho(i) = r$, so $i \in \mathcal{I}^+$ and $\varphi(i) = Pn+r+1 = k$.

Injective. If $\varphi(i) = \varphi(i')$ then $P\beta(i) + \rho(i) = P\beta(i') + \rho(i')$ with $0 \leq \rho(i), \rho(i') < P$; uniqueness of division by P forces $\beta(i) = \beta(i')$ and $\rho(i) = \rho(i')$, hence $i = i'$.

So φ is a bijection, i.e. each p_k occurs exactly once. The same argument with $\psi : \mathcal{I}^- \rightarrow \mathbb{N}$, $\psi(i) = Q\beta(i) + \rho(i) - P + 1$, and $0 \leq \rho(i) - P < Q$, handles the q_k . $\square$

The partial sums of $F(P, Q)$ are

$$S_N^{(P, Q)} = \sum_{i=1}^N b_i^{(P, Q)}, \quad N \in \mathbb{N}.$$

Define also the positive-count function

$$\pi_{P, Q}(N) = \#\{i \leq N : b_i^{(P, Q)} > 0\}.$$

Lemma 2. *Write $N = n(P+Q) + r$ with $n \in \mathbb{Z}_{\geq 0}$, $0 \leq r < P+Q$. Then*

$$\pi_{P, Q}(N) = Pn + \min(r, P),$$

and moreover the first N terms of $F(P, Q)$ consist of exactly $p_1, \dots, p_{\pi_{P, Q}(N)}$ and $q_1, \dots, q_{N-\pi_{P, Q}(N)}$, each once.

Proof. The indices $i \leq N$ are exactly those with $\beta(i) < n$, together with those with $\beta(i) = n$ and $\rho(i) < r$. Among the former, $\mathcal{I}^+$ contributes P indices per block, giving Pn . Among the latter, $\mathcal{I}^+$ contributes those with $\rho(i) < \min(r, P)$, giving $\min(r, P)$. This proves the formula.

For the second claim, by the proof of Lemma 1 the positive terms with $i \leq N$ are $p_{\varphi(i)}$ where φ is a bijection $\mathcal{I}^+ \rightarrow \mathbb{N}$ that is increasing in i (as $\varphi(i) = P\beta(i) + \rho(i) + 1$ and both β, ρ increase lexicographically with i). An increasing bijection maps the first $\pi_{P, Q}(N)$ elements of $\mathcal{I}^+$ onto $\{1, \dots, \pi_{P, Q}(N)\}$. Same for ψ and the negative terms. $\square$

3 What can and cannot be rearranged

The objection raised was that separating positive and negative terms is illegal. So before computing anything I want to fix exactly which finite rearrangements I am using, and which ones are not available with reasoning.

Different (P, Q) do not share partial sums

The natural thought: $F(P_1, Q_1)$ and $F(P_2, Q_2)$ are rearrangements of each other in the global sense (both are rearrangements of (a_j) by Lemma 1), so maybe their partial sums are rearrangements of each other too, and then they would be forced to have the same limit. This fails for large N .

Lemma 3. *Let $(P_1, Q_1), (P_2, Q_2) \in \mathbb{N} \times \mathbb{N}$ satisfy $P_1 Q_2 \neq P_2 Q_1$. Put $c_i = \frac{P_i Q_i}{P_i + Q_i}$ and*

$$N_0 = \frac{(c_1 + c_2)(P_1 + Q_1)(P_2 + Q_2)}{|P_1 Q_2 - P_2 Q_1|}.$$

Then $\forall N \in \mathbb{N}$ with $N > N_0$, the first N terms of $F(P_1, Q_1)$ and the first N terms of $F(P_2, Q_2)$ are not rearrangements of each other.

Proof. By Lemma 2, the first N terms of $F(P_i, Q_i)$ form the set

$$A_i = \{p_1, \dots, p_{\pi_i(N)}\} \cup \{q_1, \dots, q_{N-\pi_i(N)}\}, \quad \pi_i := \pi_{P_i, Q_i},$$

with no repetitions. Two finite lists without repeated entries are rearrangements of each other iff they are equal as sets. Since the p_k, q_k are pairwise distinct, $A_1 = A_2$ iff $\pi_1(N) = \pi_2(N)$. So it suffices to show $\pi_1(N) \neq \pi_2(N)$.

Put $\alpha_i = \frac{P_i}{P_i + Q_i}$ and $d_i(N) = \pi_i(N) - \alpha_i N$. Writing $N = n(P_i + Q_i) + r$ with $0 \leq r < P_i + Q_i$, Lemma 2 gives

$$d_i(N) = (P_i n + \min(r, P_i)) - (P_i n + \alpha_i r) = \min(r, P_i) - \frac{P_i r}{P_i + Q_i}.$$

Two cases. If $0 \leq r \leq P_i$, then $d_i(N) = r(1 - \alpha_i) = \frac{r Q_i}{P_i + Q_i}$, which lies in $[0, \frac{P_i Q_i}{P_i + Q_i}]$ since $r \leq P_i$.

If $P_i < r < P_i + Q_i$, then $d_i(N) = P_i - \frac{P_i r}{P_i + Q_i} = \frac{P_i(P_i + Q_i - r)}{P_i + Q_i}$, and $0 < P_i + Q_i - r < Q_i$ puts this in $(0, \frac{P_i Q_i}{P_i + Q_i})$. In both cases $|d_i(N)| \leq c_i$.

Hence, by the reverse triangle inequality,

$$|\pi_1(N) - \pi_2(N)| = |(\alpha_1 - \alpha_2)N + d_1(N) - d_2(N)| \geq |\alpha_1 - \alpha_2|N - c_1 - c_2.$$

Also

$$|\alpha_1 - \alpha_2| = \left| \frac{P_1}{P_1 + Q_1} - \frac{P_2}{P_2 + Q_2} \right| = \frac{|P_1 Q_2 - P_2 Q_1|}{(P_1 + Q_1)(P_2 + Q_2)} > 0$$

by hypothesis. So $N > N_0$ gives $|\alpha_1 - \alpha_2|N - c_1 - c_2 > 0$, whence $\pi_1(N) \neq \pi_2(N)$. $\square$

The hypothesis $P_1 Q_2 \neq P_2 Q_1$ cannot be dropped: for $(1, 1)$ and $(2, 2)$ we have $\alpha_1 = \alpha_2$ and the partial sums do agree along multiples of 4. But then $P_1/Q_1 = P_2/Q_2$ and the value computed below is the same for both, so nothing breaks.

So being a global rearrangement of each other says nothing about the partial sums, and there is no contradiction in $F(P_1, Q_1)$ and $F(P_2, Q_2)$ converging to different values.

The rearrangement I am actually using

Lemma 4. *Let $N = n(P + Q)$ for some $n \in \mathbb{N}$, i.e. $(P + Q) \mid N$. Then*

$$S_N^{(P,Q)} = \sum_{k=1}^{Pn} \frac{1}{2k-1} - \sum_{k=1}^{Qn} \frac{1}{2k}.$$

Proof. Here $r = 0$, so Lemma 2 gives $\pi_{P,Q}(N) = Pn$ and $N - \pi_{P,Q}(N) = n(P + Q) - Pn = Qn$; moreover the first N terms are exactly $p_1, \dots, p_{Pn}, q_1, \dots, q_{Qn}$, each occurring once.

Therefore $S_N^{(P,Q)}$ is a sum of the N real numbers $p_1, \dots, p_{Pn}, q_1, \dots, q_{Qn}$, and N is finite. Addition on $\mathbb{R}$ is commutative and associative, so by induction on the number of summands any permutation of a finite sum leaves it unchanged; in particular we may add all the p 's first and then all the q 's. Substituting $p_k = \frac{1}{2k-1}$ and $q_k = -\frac{1}{2k}$ gives the identity. $\square$

This is the only place where positive and negative terms are separated, it is done on a finite sum, and it happens before any limit is taken. No infinite series of positive terms and no infinite series of negative terms appears anywhere in this proof.

4 the limit along the blocks

Lemma 5. *Let $L := \ln 2 + \frac{1}{2} \ln\left(\frac{P}{Q}\right)$. Then*

$$\forall \epsilon > 0 \quad \exists n_0 \in \mathbb{N} \quad \forall n \geq n_0 : \left| S_{n(P+Q)}^{(P,Q)} - L \right| < \epsilon.$$

Proof. By Lemma 4 and Fact 1,

$$S_{n(P+Q)}^{(P,Q)} = \left(H_{2Pn} - \frac{1}{2} H_{Pn} \right) - \frac{1}{2} H_{Qn} = H_{2Pn} - \frac{1}{2} H_{Pn} - \frac{1}{2} H_{Qn}.$$

Apply (1) to each of the three harmonic numbers:

$$S_{n(P+Q)}^{(P,Q)} = (\ln(2Pn) + \gamma) - \frac{1}{2}(\ln(Pn) + \gamma) - \frac{1}{2}(\ln(Qn) + \gamma) + \delta_n,$$

where

$$\delta_n := \varepsilon_{2Pn} - \frac{1}{2}\varepsilon_{Pn} - \frac{1}{2}\varepsilon_{Qn}.$$

The γ 's cancel: $\gamma - \frac{1}{2}\gamma - \frac{1}{2}\gamma = 0$. Expanding the logarithms (all arguments are positive integers),

$$\ln 2 + \ln P + \ln n - \frac{1}{2} \ln P - \frac{1}{2} \ln n - \frac{1}{2} \ln Q - \frac{1}{2} \ln n = \ln 2 + \frac{1}{2} \ln P - \frac{1}{2} \ln Q = L,$$

so the $\ln n$ terms cancel as well and

$$S_{n(P+Q)}^{(P,Q)} = L + \delta_n \quad \forall n \in \mathbb{N}. \quad (4)$$

It remains to show $\delta_n \rightarrow 0$. Let $\epsilon > 0$. By (2) applied with $\epsilon/2$, there is $M \in \mathbb{N}$ such that $|\varepsilon_m| < \epsilon/2$ for all $m \geq M$. Put $n_0 = M$ (note $Pn \geq n$, $Qn \geq n$, $2Pn \geq n$ for all $n \in \mathbb{N}$, since $P, Q \geq 1$). Then for every $n \geq n_0$ the three indices $2Pn, Pn, Qn$ are all $\geq M$, so

$$|\delta_n| \leq |\varepsilon_{2Pn}| + \frac{1}{2} |\varepsilon_{Pn}| + \frac{1}{2} |\varepsilon_{Qn}| < \frac{\epsilon}{2} + \frac{\epsilon}{4} + \frac{\epsilon}{4} = \epsilon.$$

With (4) this is the proof for convergence for $(P + Q) \mid N$ case. $\square$

5 The general partial sum

Lemma 5 only covers N divisible by $P + Q$. We now show the remaining N change nothing.

Lemma 6. *For every $N \geq P + Q$, writing $N = n(P + Q) + r$ with $n \in \mathbb{N}$, $0 \leq r < P + Q$,*

$$\left| S_N^{(P,Q)} - S_{n(P+Q)}^{(P,Q)} \right| \leq \frac{P + Q}{2n \min(P, Q)}.$$

Proof. We have

$$S_N^{(P,Q)} - S_{n(P+Q)}^{(P,Q)} = \sum_{i=n(P+Q)+1}^N b_i^{(P,Q)},$$

a sum of $r \leq P + Q$ terms. Each such i satisfies $n(P + Q) \leq i - 1 < (n + 1)(P + Q)$, so $\beta(i) = n$ by uniqueness in the division algorithm.

By (3), if $b_i^{(P,Q)}$ is positive it equals p_k with $k = Pn + \rho(i) + 1 \geq Pn + 1$, so

$$0 < p_k = \frac{1}{2k - 1} \leq \frac{1}{2(Pn + 1) - 1} = \frac{1}{2Pn + 1} \leq \frac{1}{2Pn} \leq \frac{1}{2n \min(P, Q)}.$$

If it is negative it equals q_k with $k = Qn + \rho(i) - P + 1 \geq Qn + 1$, so

$$|q_k| = \frac{1}{2k} \leq \frac{1}{2(Qn + 1)} \leq \frac{1}{2Qn} \leq \frac{1}{2n \min(P, Q)}.$$

The triangle inequality over at most $P + Q$ such terms gives the bound. $\square$

Theorem 1. *For every $(P, Q) \in \mathbb{N} \times \mathbb{N}$ the series $F(P, Q)$ converges, and*

$$\lim_{N \rightarrow \infty} S_N^{(P,Q)} = \ln 2 + \frac{1}{2} \ln \left(\frac{P}{Q} \right).$$

Proof. Let $L = \ln 2 + \frac{1}{2} \ln(P/Q)$ and let $\epsilon > 0$.

By Lemma 5 there is $n_1 \in \mathbb{N}$ with $\left| S_{n(P+Q)}^{(P,Q)} - L \right| < \epsilon/2$ for all $n \geq n_1$.

By the archimedean property there is $n_2 \in \mathbb{N}$ with $n_2 > \frac{P + Q}{\epsilon \min(P, Q)}$, so that $\frac{P + Q}{2n \min(P, Q)} < \frac{\epsilon}{2}$ for all $n \geq n_2$.

Put $n_3 = \max(n_1, n_2)$ and $M = n_3(P + Q)$. Let $N \geq M$ and write $N = n(P + Q) + r$ with $0 \leq r < P + Q$. Then $n(P + Q) \geq N - (P + Q - 1) \geq M - (P + Q) + 1$, and since $n(P + Q)$ is a multiple of $P + Q$ that is $> (n_3 - 1)(P + Q)$, we get $n \geq n_3$.

Hence by Lemma 6 and the choice of n_1, n_2 ,

$$\left| S_N^{(P,Q)} - L \right| \leq \left| S_N^{(P,Q)} - S_{n(P+Q)}^{(P,Q)} \right| + \left| S_{n(P+Q)}^{(P,Q)} - L \right| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $S_N^{(P,Q)} \rightarrow L$. $\square$

6 conclusion

From the general result we can conclude that for the case of $F(1,1)$ we get $\ln 2$ and for $F(2,1)$ we get $\frac{3}{2} \ln 2$. Both converges. Also any of these sequence with non-zero constant P, Q converges.